

CSE 516 - PROBABILISTIC GRAPHICAL MODELS

MAJOR PROJECT

MILESTONE 1 - PROPOSAL AND SCOPING

Deep End-to-End Causal Inference:
Bayesian Structure Learning and Simulation-Based Treatment Effect Estimation under
Heterogeneous, Partially Observed Data

Track: Theoretical AI

Group: G5_TH_PGM

Project Repository: github.com/masoomchoksi/G5_TH_PGM

Base Implementation: Microsoft Causica ([microsoft/causica](https://microsoft.com/causica))

Submission date: 15 September 2026

**M1 focus: Re-derivation and boundary analysis of DECI Theorem 1, with
controlled empirical validation.**

1. Project Definition and Track

This is a Theoretical AI project. The project commits to one principal methodological question: the validity and boundary of the graph-recovery guarantee stated as Theorem 1 in Deep End-to-end Causal Inference (DECI). The empirical component is supporting evidence for the theoretical analysis, not a substitute for it.

M1 requirement	Project commitment
Selected algorithm / property	DECI causal graph-learning framework; principal focus on Theorem 1 graph-recovery guarantee.
Mathematical question	Under exactly which assumptions does DECI's ELBO optimum recover the true causal graph, and where does the proof cease to apply?
Benchmark plan	Synthetic causal graphs and controlled perturbations, with semi-synthetic causal-effect benchmarks for supporting ATE/CATE analysis.

Primary baseline	NOTEARS-MLP for nonlinear graph-recovery comparison; PC and ICALiNGAM as complementary classical comparators.
SOTA anchor	Vashishtha et al., “Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference,” ICLR 2025, with Microsoft Research affiliation.

2. Project Summary

Causal inference is a question of what ifs: what would happen to an outcome if a treatment or other variable were actually manipulated? If the causal graph is unknown, learning the causal structure, as well as inference that respects interventions, is a problem for a system. DECI addresses this by combining Bayesian causal graph learning, nonlinear structural-function learning, and simulation-based causal-effect estimation in a single framework.

The project does not aim simply to reproduce DECI. It investigates the mathematical chain behind its graph-recovery guarantee. In particular, we will re-derive the connection between maximum-likelihood recovery, ELBO maximization, the KL regularization term, and the DAG-constrained graph prior. We will then identify why the original proof does not directly transfer to mixed-type variables and partially observed data, and test whether the predicted boundaries are visible in controlled experiments.

3. Problem Statement, Motivation and Significance

1. Problem statement

Traditional causal pipelines often separate causal discovery from causal inference. A graph is first estimated from observational data and is then treated as fixed while interventions and treatment effects are computed. If the graph is wrong, downstream causal-effect estimates can inherit the structural error. DECI instead maintains uncertainty over candidate causal graphs and couples graph learning with structural-function learning and intervention simulation.

The problem studied here is not whether DECI can produce good benchmark scores in general. The focused question is whether the formal recovery guarantee can be reconstructed from its stated assumptions and, if an assumption is relaxed, what exact step of the argument fails.

2. Motivation

- Causal discovery and causal effect estimation depend on structural assumptions that should be made explicit.

- A high empirical score does not establish that a theorem applies to the data-generating regime being tested.
- DECI provides a useful end-to-end PGM case study because representation, learning and intervention inference are coupled.
- Mixed-type and missing-data settings are practically important but lie outside the direct scope of the original continuous additive-noise proof.

3. Significance

The expected value of the project is a theory-to-practice map: each theorem assumption will be linked to a model component, a proof step, and an empirical diagnostic. This makes the project more than an implementation exercise and directly answers the course question of where the PGM reasoning is.

4. Research Questions and Hypothesis

1. Can DECI Theorem 1 be rigorously re-derived from its stated assumptions?
2. In the limit of infinite data, what does maximum likelihood recovery of the ground-truth joint distribution contribute to the graph-recovery argument?
3. What is the relation between convergence of the ELBO-maximizing variational posterior and the MLE solution, and what does the KL term and DAG-constrained prior do?
4. Which exact theorem assumptions are violated by mixed-type/discrete variables, and what proof step is no longer justified?
5. Why does the missing-data extension, which uses an approximate amortized imputation posterior, fall outside the original proof?
6. How do these theoretical boundaries manifest empirically in graph recovery and ATE/CATE estimation?

Working hypothesis

When the theorem's continuous additive-noise, identifiability, no-latent-confounding, correct-specification and regularity conditions are satisfied, DECI should exhibit improved graph-recovery behavior as sample size increases. When those assumptions are violated, graph posterior uncertainty and structural error should increase, and downstream ATE/CATE error should degrade. This is a hypothesis to be tested, not a claimed M1 result.

5. Mathematical Formulation

1-Causal graph

Assume that the causal structure is a directed acyclic graph (DAG) $G = (V, E)$, with each node as a random variable and a directed edge $X_i \rightarrow X_j$ as a direct causal dependency between random variables X_i and X_j .

2-Structural equation model

For each variable X_i , the nonlinear additive-noise SEM is:

$$X_i = f_i(X_{pa(i)}) + Z_i$$

Here $X_{pa(i)}$ is the parent set of X_i , f_i is a nonlinear structural function, and Z_i is an exogenous noise variable. The project will explicitly track which theorem assumptions are required for identifiability of the causal structure.

3-Factorization and conditional independence

Under the DAG Markov assumption, the joint distribution factorizes according to the graph:

$$p(x_1, \dots, x_d | G, \theta) = \prod_i p(x_i | x_{pa(i)}, \theta_i)$$

The graph also encodes conditional-independence statements through d-separation. Thus, a missing edge is not merely a visual absence: together with the Markov assumptions, it asserts conditional independencies implied by the graph. The project will examine how incorrect or non-identifiable structures affect these implied dependencies.

4-Bayesian structure learning

DECI treats G as uncertain rather than fixed. Conceptually, the target is a posterior $q\phi(G)$ over DAGs, obtained by optimizing a variational objective involving the data likelihood, a DAG-constrained prior and a KL term. The project will reconstruct how the ELBO objective connects to the MLE solution in the large-data limit.

5-Intervention and causal effect

For treatment X_T and outcome XY , the intervention $do(X_T = a)$ replaces the treatment-generating mechanism with the chosen value. The Average Treatment Effect is:

$$ATE(a, b) = E[XY | do(X_T = a)] - E[XY | do(X_T = b)]$$

CATE conditions the treatment effect on relevant covariates. DECI estimates these quantities by simulating the learned structural equations under intervention while accounting for uncertainty over candidate graphs.

6. Mandatory PGM Mapping: Representation, Learning and Inference

Vertical	What the project represents / computes	Why appropriate
Representation	Random variables X_i ; DAG nodes; directed causal edges; parent sets; nonlinear SEMs; exogenous noise; graph posterior $q\phi(G)$.	A DAG makes causal dependencies and conditional-independence assumptions explicit and supports intervention semantics.
Learning	Bayesian structure learning plus functional/parameter learning. Optimize an ELBO using a DAG-constrained prior and learn nonlinear structural mechanisms/noise distributions.	The graph is unknown, so structure learning is necessary; the conditional mechanisms are also unknown, so parameter/function learning is coupled to graph learning.
Inference	Posterior over graphs $\rightarrow$ intervention $\text{do}(X_T=a) \rightarrow$ structural simulation $\rightarrow$ outcome distribution $\rightarrow$ ATE/CATE; uncertainty is marginalized over graph samples.	Exact inference is generally impractical for nonlinear flow-based mechanisms and graph uncertainty; simulation provides an intervention-aware route to causal quantities.

1. Conditional-independence assumptions

- The DAG is assumed to encode the relevant causal structure under the stated causal-sufficiency setting.
- The distribution is Markov with respect to the DAG, so d-separation corresponds to conditional independence under the model assumptions.
- No latent confounders are assumed in the theorem setting; the project will distinguish this formal assumption from more realistic partially observed experiments.

2. Representation $\rightarrow$ Learning $\rightarrow$ Inference chain

Observational data $\rightarrow$ Bayesian DAG posterior $\rightarrow$ nonlinear SEM learning $\rightarrow$ intervention simulation $\rightarrow$ ATE/CATE

7. Theoretical Focus: DECI Theorem 1

The central theoretical task is to reconstruct the proof rather than cite the theorem as a black box. The analysis will be organized as follows.

Proof component	Question to be answered	Planned validation
MLE recovery	Under the model and identifiability assumptions, why does maximum likelihood recover the ground-truth joint distribution and causal graph in the infinite-data limit?	Formal derivation and controlled increasing-sample experiments.
ELBO-to-MLE convergence	Why does an optimal variational solution approach the MLE solution as data grow?	Track each optimization/objective term and reproduce the supporting convergence argument.
KL and DAG prior	How does the KL regularization and differentiable DAG-constrained prior affect the variational posterior over graphs?	Ablation/diagnostic analysis where computationally feasible.
Boundary: mixed type	Which continuous additive-noise/identifiability step no longer transfers directly to mixed discrete-continuous variables?	Compare continuous and mixed-type settings.
Boundary: missing data	How does approximate amortized imputation change the objective and invalidate a direct application of the original proof?	Vary missingness rate and report graph/ATE-CATE degradation.

1 Theorem assumptions to track

Assumption	Role	Failure mode to study
Correct specification	True data-generating process lies in the model family.	Model mismatch can prevent exact recovery.
No latent confounding / causal sufficiency	Relevant common causes are observed.	Unobserved confounding can make the learned graph ambiguous.

Structural identifiability	Distribution identifies a unique causal structure in the stated class.	Non-identifiable graphs can yield posterior uncertainty.
Continuous additive-noise setting	Supports the original nonlinear ANM proof.	Mixed/discrete variables require a different argument.
Non-trivial, sufficiently regular functions	Supports identifiability and proof steps.	Degenerate or non-regular functions can break the argument.
Regular likelihood/noise density	Supports likelihood-based convergence.	Irregular likelihoods may invalidate the convergence argument.

8. State-of-the-Art Positioning

The course guidelines require a specific recent work from a major technology company or international research laboratory, with explicit analysis of its problem, representation, learning/inference methodology, evaluation, findings, limitations and relationship to the project. The following paper is used as the M1 SOTA anchor.

1 SOTA anchor: Microsoft Research / ICLR 2025

Vashishtha, A., Reddy, A. G., Kumar, A., Bachu, S., Balasubramanian, V. N., & Sharma, A. “Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference.” ICLR 2025. The paper has Microsoft Research, India affiliations and studies how imperfect human/LLM experts can provide causal information.

Required SOTA element	Analysis
Problem	Pairwise expert judgments can confuse direct and indirect effects and can produce cycles, so the paper seeks a more stable causal-order interface.
Representation	The central structural object is a causal order rather than directly trusting a graph assembled from pairwise expert judgments; auxiliary variables are used to make local expert queries more cycle-consistent.

Learning / methodology	The approach uses prompting/query design to obtain causal-order information from imperfect experts, then uses that information in downstream causal discovery/inference.
Inference / downstream use	The estimated causal order is used to reduce errors in downstream discovery and effect-inference tasks.
Evaluation	The paper evaluates cycle formation, causal-order accuracy and downstream effects across real-world graphs using LLMs and human annotators; it also studies robustness to different auxiliary variables and model sizes.
Major findings	Triplet-style prompting produces fewer cycles and more accurate orders than pairwise prompting; the paper reports robustness gains and downstream improvements.
Limitations / distinction	The paper relies on expert/LLM information and focuses on causal-order extraction, whereas our project studies observational-data Bayesian DAG learning and the mathematical recovery guarantee of DECI.
Relationship to our project	It is complementary SOTA: it shows a current research direction for improving causal discovery under imperfect structural information. Our project asks a different but closely related question—when an end-to-end probabilistic causal learner has a formal graph-recovery guarantee and where that guarantee breaks.

2 Foundational method being analyzed: DECI

The principal method is Geffner et al.'s Deep End-to-end Causal Inference. Microsoft Research describes DECI as a single flow-based method that performs both causal discovery and causal inference, including CATE estimation, and reports a theoretical recovery guarantee under stated assumptions. This is the method under investigation, not the separate recent SOTA anchor.

3 M1 SOTA positioning statement

At M1, the project is positioned as a theoretical audit and boundary analysis of an end-to-end Bayesian causal framework, informed by recent causal-discovery research. We do not claim to outperform the SOTA anchor. The later milestones will use empirical evidence to reposition the project based on graph recovery, uncertainty, causal-effect estimation and computational cost.

9. Benchmark and Dataset Plan

Because this is Theoretical AI, the project is not tied to one application domain. The benchmarks are chosen to isolate the mathematical assumptions under study and then to test whether the same boundaries affect causal-effect estimation.

Benchmark	Role in project	Primary outputs
CSuite	Controlled synthetic causal cases including non-identifiability, non-Gaussian noise, Simpson's paradox, adjustment-set difficulty and mixed variables.	SHD, precision/recall/F1, graph posterior uncertainty.
ER / Scale-Free random DAGs	Scalability and recovery under controlled graph size and density.	SHD/F1/runtime vs sample size, variables and edges.
SynTReN	Biologically motivated synthetic regulatory-network benchmark.	Structure-recovery metrics and robustness.
Sachs	Real-data causal discovery benchmark with an expert-curated graph.	Structural recovery and qualitative graph comparison.
IHDP	Semi-synthetic causal-effect benchmark.	ATE/CATE RMSE and uncertainty.
Twins	Semi-synthetic causal-effect benchmark.	ATE/CATE error and robustness.

10. Baselines and Comparative Methods

The primary baseline for nonlinear graph recovery will be NOTEARS-MLP because it provides a continuous-optimization alternative for nonlinear DAG learning. PC and ICALiNGAM will serve as complementary classical causal-discovery references. For treatment-effect estimation, established DoWhy-based linear and nonlinear DML pipelines will be used where compatible with the benchmark.

Task	Primary / comparator	Purpose
Graph discovery	DECI (main) vs NOTEARS-MLP (primary baseline)	Compare nonlinear structural recovery under matched benchmark conditions.

Graph discovery	PC	Classical constraint-based reference; highlights differences in assumptions.
Graph discovery	ICALiNGAM	Linear/non-Gaussian causal discovery reference.
Causal inference	DECI vs DoWhy-Linear / DoWhy-Nonlinear DML	Compare ATE/CATE estimation under compatible evaluation settings.

All comparisons will document assumptions and keep dataset splits, sample sizes, hardware, random seeds and evaluation procedures as consistent as possible. The goal is not to assume that DECI must win, but to identify regimes in which its probabilistic graph treatment and end-to-end coupling provide an advantage or incur a cost.

11. Initial KPIs

The KPIs are deliberately tied to Representation, Learning and Inference. Accuracy alone will not be treated as sufficient evidence.

PGM stage	KPI	Interpretation	Direction
Representation	SHD	Graph-edit distance between learned and ground-truth DAG.	↓
Representation / Learning	Edge Precision	Fraction of predicted edges that are correct.	↑
Representation / Learning	Edge Recall	Fraction of true edges recovered.	↑
Representation / Learning	F1	Overall structural recovery.	↑
Representation	Posterior graph uncertainty	Uncertainty over candidate DAGs, especially in non-identifiable regimes.	Interpret
Learning	Log-likelihood / NLL	Compatibility of learned probabilistic model with data.	↑ / ↓
Learning	ELBO	Optimization objective linking likelihood and variational posterior.	↑

Learning	Parameter/function estimation error	Accuracy of learned structural mechanisms where ground truth is available.	↓
Inference	ATE RMSE	Error in estimated average treatment effects.	↓
Inference	CATE RMSE	Error in heterogeneous/conditional treatment effects.	↓
Inference	Inference runtime	Computational cost of causal-effect estimation.	↓
Inference	Monte Carlo / posterior variability	Sensitivity of effect estimates to graph uncertainty and sampling.	Interpret

Planned plots include Sample Size vs SHD/F1, Graph Density vs SHD, Missingness vs ATE/CATE RMSE, and Inference Error vs Runtime. Repeated runs will report mean $\pm$ standard deviation and, where appropriate, 95% confidence intervals.

12. Experimental Plan

Experiment	Independent factor	Primary analysis
E1 - Sample size	Number of observations	Does graph recovery improve toward the theorem prediction as n increases?
E2 -Graph size	Number of variables	Recovery, F1, runtime and scalability.
E3 - Graph density	Number of edges	Recovery degradation as dependency complexity increases.
E4 -Noise model	Gaussian vs non-Gaussian / spline noise	Sensitivity to the theorem's noise assumptions.
E5 - Identifiability	Identifiable vs non-identifiable cases	Posterior uncertainty and structural recovery.
E6 -Variable type	Continuous vs mixed/discrete	Empirical behavior at the theoretical boundary.
E7 - Missingness	Missing-data rate and imputation setting	Graph and ATE/CATE degradation; distinguish empirical behavior from formal guarantees.
E8 - Discovery benchmark	CSuite, ER/SF, SynTReN, Sachs	DECI vs discovery baselines.

E9 - Causal inference	IHDP, Twins and selected controlled settings	ATE/CATE RMSE and comparative ranking.
E10 - Theory-to-practice	Assumption satisfied vs violated	Check whether observed behavior agrees with theoretical predictions.

1 Statistical analysis plan

- Use multiple random seeds / repeated runs where computationally feasible.
- Report mean $\pm$ standard deviation rather than a single run.
- Use 95% confidence intervals where appropriate.
- For paired method comparisons, use paired tests or bootstrap confidence intervals when the experimental design supports them.
- Interpret effect sizes and variability; do not treat statistical testing as a substitute for PGM reasoning.

13. Scope, Delimitations and Expected Contribution

1 In scope

- DECI's Bayesian causal graph-learning and simulation-based inference pipeline.
- Theorem 1 assumptions, proof structure and ELBO-to-MLE argument.
- Two targeted theoretical boundaries: mixed-type variables and missing observations.
- Controlled empirical validation of graph recovery and causal-effect behavior.

2 Out of scope

- A claim of a new universal causal-identification theorem without a complete proof.
- A claim that benchmark performance alone establishes theoretical validity.
- Broad coverage of unrelated causal algorithms.
- Treating empirical mixed-type or missing-data performance as a formal theorem extension unless a complete proof is established.

3 Expected contribution

The expected contribution is a rigorous, self-contained explanation of the scope of DECI's recovery guarantee and a controlled theory-to-practice analysis showing how assumption satisfaction or violation affects graph recovery and causal-effect estimation.

14. Implementation and Technical Artefacts

- Derivation notes and proof reconstruction for Theorem 1.
- DECI/Causica experiment configurations and scripts.
- Synthetic benchmark generation/preparation.
- Baseline execution for PC, NOTEARS variants and ICALiNGAM.
- Simulation-based ATE/CATE estimation pipeline.
- Automated KPI calculation and plotting scripts.
- Dataset documentation and reproducibility instructions.

15. Team Responsibilities

The course guidelines require each workstream to be assigned to actual group members before submission. The original report left this as a placeholder; the table below makes the required ownership explicit. Replace the bracketed names with the official team members before uploading.

Workstream	Assigned member	Responsibility
Theory / proof	Parisha Kathiria	Re-derive Theorem 1, document assumptions, MLE recovery, ELBO convergence and KL/DAG-prior argument.
Gap analysis	Vrunda Patel	Analyze mixed-type and missing-data extensions and identify proof-breaking points.
DECI implementation	Yashvi Patel	Set up and document Microsoft Causica implementation and experiment configurations.
Baselines	Masoom Choksi	Configure/run PC, NOTEARS variants, GraN-DAG/ICALiNGAM and causal-inference baselines as scheduled.
Datasets & experiments	Khushi Thakar	Prepare CSuite, ER/SF, SynTReN, Sachs, IHDP and Twins experiments.
Evaluation & visualization	ALL members	Compute discovery/inference KPIs, produce plots/tables, and connect results to theoretical predictions.

16. Risks, Limitations and Mitigation

Risk / limitation	Mitigation
Proof assumptions may be too strong for an extension.	State the exact boundary rather than over-claiming; identify additional assumptions or model changes needed.
DECI experiments may be computationally expensive.	Start with small controlled benchmarks and scale systematically.
Some settings are non-identifiable.	Treat them as explicit uncertainty experiments rather than as simple theorem violations.
Missing-data behavior depends on imputation.	Report missingness rate and imputation configuration and separate empirical observations from formal guarantees.
Baselines use different assumptions.	Document assumptions and interpret comparisons within compatible regimes.
Team contribution may be difficult to demonstrate.	Assign named workstreams, preserve derivations/code ownership and ensure every member can explain the PGM components.

17. Reproducibility Plan

- Keep report, derivations, source code, experiment configurations, dataset documentation, scripts and generated results clearly separated in GitHub.
- Record environment/software versions and commands needed to reproduce experiments.
- Use explicit configurations and controlled random seeds where feasible.
- Store result-generation scripts so that tables and plots can be regenerated.
- Document benchmark download/preparation steps rather than relying on undocumented local files.

18. Conclusion

This M1 proposal defines a focused Theoretical AI investigation of DECI centered on Theorem 1. The project maps the method onto the PGM representation–learning–inference framework, makes its factorization and conditional-independence assumptions explicit, identifies the mathematical recovery question, specifies baselines and PGM-relevant KPIs, and provides a controlled experimental plan for testing theoretical boundaries. The SOTA position is anchored

to a specific recent Microsoft Research/ICLR 2025 causal-inference paper rather than to an organisation name alone.

The intended outcome is not simply a performance comparison. It is a technically defensible explanation of how DECI's probabilistic representation and variational learning objective lead to its causal graph-recovery guarantee, and how the assumptions behind that guarantee relate to observed behavior when the data regime changes.

19. References

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